39.	All of the follwing are suitable problem for genetic algorithm EXCEPT
A.	pattern recognization
B.	simulation of biological models
C.	simple optimization with few variables
D.	dynamic process control

41.	Genetic algorithms are example of
A.	heuristic
B.	evolutionary algorithm
C.	aco
D.	pso

Show answer

discuss

42.	mutation is applied on __candidates.
A.	one
B.	two
C.	more than two
D.	noneof these

Show answer

discuss

43.	recombination is applied on __candidates.
A.	one
B.	two
C.	more than two
D.	noneof these

Show answer

discuss

44.	LCS belongs to ___ based methods?
A.	rule based learning
B.	genetic learning
C.	both a and b
D.	noneof these

Show answer

discuss

45.	Survival is ___ approach.
A.	deteministic
B.	non deterministic

45.	**Survival is ___ approach.**
C.	semi deterministic
D.	noneof these

Show answer

discuss

46.	**Evolutionary algorithms are a ___ based approach**
A.	heuristic
B.	metaheuristic
C.	both a and b
D.	noneof these

Show answer

discuss

47.	**Idea of genetic algorithm came from**
A.	machines
B.	birds
C.	aco
D.	genetics

Show answer

discuss

48.	**Chromosomes are actually ?**
A.	line representation
B.	string representation
C.	circular representation
D.	all of these

Show answer

discuss

49.	**what are the parameters that affect GA are/is**
A.	selection process
B.	initial population
C.	both a and b
D.	none of these

Show answer

discuss

50.	**Evolutionary programming was developef by**
A.	fredrik

50.	**Evolutionary programming was developef by**
B.	fodgel
C.	frank
D.	flin

Show answer

56.	**What are different types of crossover**
A.	discrete and intermedium
B.	discrete and continuous
C.	continuous and intemedium
D.	none of these

Show answer

discuss

57.	**Determining the duration of the simulation occurs before the model is v**
A.	true
B.	false

Show answer

discuss

67.	**Parameters that affect GA**
A.	initial population
B.	selection process
C.	fitness function
D.	all of these

Show answer

discuss

68.	**Fitness function should be**
A.	maximum
B.	minimum
C.	intermediate
D.	noneof these

Show answer

discuss

69.	**Applying recombination and mutation leads to a set of new candidates,**
A.	sub parents
B.	parents
C.	offsprings

69.	**Applying recombination and mutation leads to a set of new candidates, (**
D.	grand child

Show answer

discuss

70.	**_____ decides who becomes parents and how many children the parents h**
A.	parent combination
B.	parent selection
C.	parent mutation
D.	parent replace

Show answer

discuss

71.	**Basic elements of EA are ?**
A.	parent selection methods
B.	survival selection methods
C.	both a and b
D.	noneof these

Show answer

discuss

102.	**In Evolutionary programming, survival selection is**
A.	probabilistic selection (μ+μ) selection
B.	(μ, λ)- selection based on the children only (μ+λ)- selection based on both the set of parent
C.	children replace the parent
D.	all the mentioned

Answer» A. probabilistic selection (μ+μ) selection

104.	**In Evolutionary programming, recombination is**
A.	doesnot use recombination to produce offspring. it only uses mutation
B.	uses recombination such as cross over to produce offspring
C.	uses various recombination operators
D.	none of the mentioned

Answer» A. doesnot use recombination to produce offspring. it only uses mutation

111.	**In Evolutionary programming,**
A.	individuals are represented by real-valued vector

102. **In Evolutionary programming, survival selection is**

B. individual solution is represented as a finite state machine

C. individuals are represented as binary string

D. none of the mentioned

Answer» B. individual solution is represented as a finite state machine

15. **Termination condition for EA**

A. mazimally allowed cpu time is elapsed

B. total number of fitness evaluations reaches a given limit

C. population diveristy drops under a given threshold

D. all the mentioned

Answer» D. all the mentioned

discuss

116. **Which of the following operator is simplest selection operator?**

A. random selection

B. proportional selection

C. tournament selection

D. none

Answer» A. random selection

discuss

117. **Which crossover operators are used in evolutionary programming?**

A. single point crossover

B. two point crossover

C. uniform crossover

D. evolutionary programming doesnot use crossover operators

Answer» D. evolutionary programming doesnot use crossover operators

discuss

118. **(1+1) ES**

A. operates on population size of two

B. operates on populantion size of one

C. operates on populantion size of zero

D. operates on populantion size of λ

Answer» A. operates on population size of two

discuss

119. **Which of these emphasize of development of behavioral models?**

A. evolutionary programming

B. genetic programming

C. genetic algorithm

D. all the mentioned

Answer» A. evolutionary programming

discuss

120. **EP applies which evolutionary operators?**

A.	variation through application of mutation operators
B.	selection
C.	both a and b
D.	none of the mentioned

Answer» C. both a and b

discuss

121. **Which selection strategy works with negative fitness value?**

A.	roulette wheel selection
B.	stochastic universal sampling
C.	tournament selection
D.	rank selection

Answer» D. rank selection

discuss

155. **categories of EA are/is**

A.	genetic algorithm
B.	genetic programing
C.	learning classifier systems
D.	all of these

Answer» D. all of these

discuss

158. **Genetic algorithm is a subset of_______.**

A.	evolutionary algorithm
B.	dynamcic algorithm
C.	both a&b
D.	none of these

Answer» A. evolutionary algorithm

discuss

159. **NP hard problems are also called as________.**

A.	dicrete optimization
B.	combinatorial optimization
C.	evolutionary optimization
D.	none of these

Answer» B. combinatorial optimization

discuss

160. **Genetic algorithm is first introduce by_______.**

A.	charles darwin
B.	john holland
C.	gregor johan mendel
D.	none of these

160.	**Genetic algorithm is first introduce by_______.**

Answer» B. john holland

discuss

161.	**___________ replicates the most successful solutions found in a populatio**
A.	selection
B.	recombination
C.	mutation
D.	none of these

Answer» A. selection

discuss

162.	**__________ decomposes two distinct solutions and then randomly mixes**
A.	selection
B.	recombination
C.	mutation
D.	none of these

Answer» B. recombination

discuss

163.	**___________ randomly perturbs a candidate solution.**
A.	selection
B.	recombination
C.	mutation
D.	none of these

Answer» C. mutation

discuss

164.	**A _________ is a template consisting of a string composed of three symbo**
A.	wild symbol
B.	schema
C.	layout
D.	none of these

Answer» B. schema

discuss

166.	**Metaheuristics are ?1)non deterministic 2)non approximate 3)not probl**
A.	1,2,3
B.	1,2
C.	1,3
D.	2,3

Answer» C. 1,3

discuss (1)

167.	**In search techniques, as single point based contradicts population based**
A.	stochastic
B.	simplex based

167.	**In search techniques, as single point based contradicts population based**
C.	complex based
D.	none

Answer» A. stochastic

discuss

168.	**In swarm systems organisations are**
A.	centalized
B.	decentralized
C.	controlled by third party
D.	none

Answer» B. decentralized

discuss

169.	**Identify the working sequence of kmean clustering ? 1)redefine cluster c clusters near centroids**
A.	1,3,2
B.	3,2,1
C.	2,3,1
D.	2,1,3

Answer» C. 2,3,1

discuss

170.	**Every particle in the system takes experience from previous particle ?**
A.	pso
B.	aco
C.	clustering
D.	none

Answer» A. pso

discuss

171.	**swarm intelligence includes ? 1)bee colony algorithm 2)ant colony algor**
A.	1,2
B.	1,2,3
C.	2,3,4
D.	all of these

Answer» D. all of these

discuss

172.	**pheromone quantity in ACO is ___ proportional to path selection.**
A.	directly
B.	inversly
C.	there is no connection
D.	none

Answer» A. directly

discuss

173.	The ants prefer the smaller drop of honey over the more abundant, but of?
A.	kruskal algorithm
B.	travelling salesman
C.	knapsack problem
D.	np hard problem

Answer» C. knapsack problem

discuss

174.	In kmeans clustering each cluster is associated with
A.	centroid
B.	edge
C.	common point
D.	none of them

Answer» A. centroid

discuss

175.	What is EC?
A.	computer based problem solving systems
B.	systems that uses computational models of evolutionary process
C.	both a and b
D.	none of these

Answer» C. both a and b

discuss

176.	Recombination is applied to
A.	2 selected candidated
B.	1 selected candidate
C.	3 selected candidate
D.	none of these

Answer» A. 2 selected candidated

discuss

177.	In EA mutation is applied to
A.	2 candidate
B.	1 candidate
C.	3 candidate
D.	none of these

Answer» B. 1 candidate

discuss

180.	GA stands for
A.	genetic algorithm
B.	genetic programing
C.	genetic assurance
D.	none of these

180. **GA stands for**

Answer» A. genetic algorithm

discuss

181. **Features of GA**

A. a string representation of chromosomes.
B. a fitness function be to minimized.
C. a cross-over method and a mutation method.
D. all of these

Answer» D. all of these

discuss

182. **GP individual stores computer program**

A. true
B. false

Answer» A. true

discuss

183. **GP selection is**

A. deterministic selection
B. tournament selection
C. nondeterministic selection
D. none of these

Answer» B. tournament selection

discuss

229. **In the generational model of EA**

A. entire set of population is replaced by offsprings
B. one member of population is replaced
C. no population member is replaced
D. depends on fitness value

Answer» A. entire set of population is replaced by offsprings

discuss

230. **In the steady-state model of EA**

A. entire set of population is replaced by offsprings
B. one member of population is replaced
C. no population member is replaced
D. depends on fitness value

Answer» B. one member of population is replaced

discuss

231. **In the generational model of EA**

A. each individual survives exactly for two generation
B. each individual survives exactly for one generation
C. cannot predict
D. each individual survives as many generations as want

231. **In the generational model of EA**

Answer» B. each individual survives exactly for one generation

discuss

232. **In the steady-state model of EA**

A.	one offspring is generated per generation
B.	two offsprings are generated per generation
C.	cannot decide
D.	more than two offsprings are generated per generation

Answer» A. one offspring is generated per generation

discuss

233. **Which of the following algorithm is most efficient for discotinuous and r**

A.	evolutionary algorithm
B.	classical optimization algorithm
C.	genetic algorithm
D.	none

Answer» A. evolutionary algorithm

discuss

234. **Each iteration of EA is referred to as**

A.	generation
B.	iteration
C.	population
D.	none

Answer» A. generation

discuss

235. **Which of the following are evolutionary operators?**

A.	selection
B.	crossover
C.	mutation
D.	all of the above

Answer» D. all of the above

discuss

240. **Fitness proportionate selection methods are**

A.	roulette wheel selection
B.	stochastic universal sampling
C.	tournament selection
D.	all the mentioned

Answer» D. all the mentioned

discuss

241. **In which selection method of survival selection there is no notion of fitn**

A.	fitness based selection
B.	elitism

241. **In which selection method of survival selection there is no notion of fitn**

C. agebased selection

D. all the mentioned

Answer» C. agebased selection

discuss

242. **In which selection strategy every individual has the same probability to**

A. roulette wheel selection

B. uniform selection

C. tournament selection

D. rank selection

Answer» B. uniform selection

discuss

243. **High selection pressure is desirable, when we need**

A. diversity not found in each generation

B. there is no improvement in successive ga iteration

C. faster termination of ga

D. fitness values are not uniformly distributed

Answer» C. faster termination of ga

discuss

244. **Tournament selection scheme is more preferable when**

A. when fitness values are uniformly distributed

B. population are with very diversified fitness values

C. when fitness values are not necessarily uniformly distributed

D. under all the above situations

Answer» B. population are with very diversified fitness values

discuss

245. **Which of the following is not a characterstic of evolutionary algorithm?**

A. conceptual simplicity

B. parallelism

C. broad applicability

D. artificial selection

Answer» D. artificial selection

discuss

246. **What is the correct order of steps in evolutionary algorithm?**

A. select parents-recombine-mutate-evaluate

B. select parents-recombine-evaluate-mutate-

C. select parents-mutate - recombine-evaluate

D. select parents-evaluate -recombine-mutate

Answer» A. select parents-recombine-mutate-evaluate

discuss

247.	**Which of the following schemes are selection schemes in Evolutionary c**
A.	hall of fame
B.	rank based selection
C.	tournament selection
D.	all of the above

Answer» D. all of the above

discuss

248.	**To encode chromosomes which encoding schemes are used**
A.	binary encoding
B.	finite state machine encoding
C.	real value encoding
D.	all of the above

Answer» D. all of the above

discuss

265.	**where does PSO based clustering terminates?**
A.	global optimum
B.	local optimum
C.	global maximum
D.	local minimum

Answer» A. global optimum

discuss

266.	**Each particle in PSO modifies its position according to ? 1)its velocity 2**
A.	2,3
B.	1,2
C.	1,2,3
D.	only 1

Answer» C. 1,2,3

discuss

267.	**Applications of ACO are ? 1)shortest path 2)assignment problem 3)set p**
A.	2,3
B.	1,2
C.	1,2,3

Answer» C. 1,2,3

discuss

268.	**Evaporation of pharomones is ?**
A.	directly proportional to path length
B.	inversly proportional to path length
C.	constant
D.	none

Answer» A. directly proportional to path length

discuss

269.	Metaheuristics doesnot include ?
A.	evolutinary algorithms
B.	tabu searching
C.	simulated annealing
D.	none

Answer» D. none

discuss

270.	Recombinbation involves __candidates while mutation requires__ candi
A.	1,2,offspring
B.	2,1,offspring
C.	1,2,parent
D.	2,1,parent

Answer» A. 1,2,offspring

discuss

271.	Identify the correct sequence for evoltionary algorithms ? 1)Select genit newborn offspring 3)Create offspring 4)replace some parents
A.	1,2,3,4
B.	1,3,2,4
C.	2,1,3,4
D.	2,3,1,4

Answer» B. 1,3,2,4

discuss

276.	Parameters that affect GA are ? 1)intial population 2)fitness func
A.	1,3
B.	1,2
C.	1,2,3
D.	only 1

Answer» B. 1,2

discuss

278.	**Identify the algorithm ?procedure EA
	{
	t = 0;
	initialize population P(t);
	evaluate P(t);
	until (done) {
	t = t + 1;
	parent_selection P(t);
	recombine P(t);
	mutate P(t);
	evaluate P(t);
	survive P(t);
	}
	}**

A.	aco
B.	pso
C.	ea
D.	kmean

Answer» C. ea

discuss

279.	**Genetic algorithm are heuristic methods that do not guarantee an optim**
A.	true
B.	false

Answer» A. true

discuss

281.	**Evolution strategies uses ? 1)selection 2)recombination 3)real val**
A.	1,2,3
B.	2,3
C.	1,3
D.	only 2

Answer» A. 1,2,3

discuss

283.	**Evolutionary Algorithms includes ? 1) tabu search 2)simulated annealin**
A.	1,2,3,4
B.	2,3
C.	3,4
D.	2,3,4

283. **Evolutionary Algorithms includes ? 1) tabu search 2)simulated annealin**

Answer» C. 3,4

discuss

284. **Which of the following statements are true ? 1)mutation is applied to tw and results in one or two new candidates, the children 2) recombination one new candidate.**

A. only 1

B. only 2

C. both

D. none of these

Answer» D. none of these

discuss

285. **The ___ is performed by selecting two sub trees by chance from the parents and exchange them to create the descendants.**

A. gp crossover

B. gp mutation

C. gp clustering

D. gp recombination

Answer» A. gp crossover

discuss

286. **The____ is performed by selecting a sub tree of the descendant by chan arbitrary generated new sub tree.**

A. gp crossover

B. gp mutation

C. gp clustering

D. gp recombination

Answer» B. gp mutation

discuss

307. **Sequence of steps in EA**

A. initialization-> selection- >mutation- >crossover- >termination

B. initialization-> selection- >crossover- >termination

C. initialization-> selection->crossover- >mutation- >termination

D. None of these

Answer» C. initialization-> selection->crossover- >mutation- >termination

discuss

309. **Which of the following is not true for Genetic algorithms?**

A. It is a probabilistic search algorithm

B. It is guaranteed to give global optimum solutions

309.	**Which of the following is not true for Genetic algorithms?**
C.	If an optimization problem has more than one solution, then it will return all the solutions
D.	It is an iterative process suitable for parallel programming

Answer» B. It is guaranteed to give global optimum solutions

discuss

310.	**Which one of the following is not necessarily be considered as GA paran**
A.	the population size.
B.	the obtainable accuracy
C.	the mutation probability
D.	the average fitness score

Answer» D. the average fitness score

discuss

311.	**Which of the following optimization problem(s) can be better solved wit**
A.	0-1 Knapsack problem
B.	Travelling salesman problem
C.	Job shop scheduling problem
D.	Optimal binary search tree construction problem

Answer» B. Travelling salesman problem

discuss

312.	**If crossover between chromosomes in search space does not produce sig imply? (if offspring consist of one half of each parent) (i) The crossover operation is not successful. (ii) Solution is about to be reached. (iii) Diversity is so poor that the parents involved in the crossover opera (iv) The search space of the problem is not ideal for GAs to operate**
A.	ii, iii & iv only
B.	ii, iii only
C.	i, iii & iv only
D.	All of the mentioned

Answer» B. ii, iii only

discuss

313.	**In Rank-based selection scheme, which of the following is not correct**
A.	The % area to be occupied by an individual , is given by average of sumation of elements
B.	Two or more individuals with the same fitness values should have the same rank.
C.	Individuals are arranged in a descending order of their fitness values.
D.	The proportionate based selection scheme is followed based on the assigned rank.

Answer» C. Individuals are arranged in a descending order of their fitness values.

discuss

314.	**Real Coded GA flow is-**
A.	Random mutation- Polynomial mutation

314. Real Coded GA flow is-

B. Polynomial mutation-Random mutation
C. Flipping-Random mutation- Polynomial mutation
D. None

Answer» A. Random mutation- Polynomial mutation

discuss

315. Breeding in GA flow is-

A. Create a mating pool- Select a pair- Reproduce
B. Select a pair-Create a mating pool- Reproduce
C. Reproduce-Create a mating pool- Select a pair
D. None

Answer» A. Create a mating pool- Select a pair- Reproduce

discuss

318. Premature convergence of PSO is

A. Once PSO traps in global optimum, it is dificult to jump out of global optimum
B. Once PSO traps in local optimum, it is dificult to jump out of local optimum
C. Once PSO traps in local optimum, it is dificult to jump out of global optimum
D. Once PSO traps in global optimum, it is dificult to jump out of local optimum

Answer» B. Once PSO traps in local optimum, it is dificult to jump out of local optimum

discuss

342. All of the follwing are suitable problem for genetic algorithm EXCEPT

A. pattern recognization
B. simulation of biol
C. simple optimization
D. dynamic process contr

Answer» C. simple optimization

discuss

343. Tabu search is an example of ?

A. heuristic
B. Evolutionary algo
C. ACO
D. PSO

Answer» A. heuristic

discuss

344. Genetic algorithms are example of

A. heuristic
B. Evolutionary algo
C. ACO
D. PSO

Answer» B. Evolutionary algo

discuss

345. mutation is applied on candidates.

A. one
B. two
C. more than two
D. noneof these

Answer» A. one

discuss

346. recombination is applied on candidates.

A. one
B. two
C. more than two
D. noneof these

Answer» B. two

discuss

347. LCS belongs to based methods?

A. rule based learning
B. genetic learning
C. both a and b
D. noneof these

Answer» A. rule based learning

discuss

348. Survival is approach.

A. deteministic
B. non deterministic
C. semi deterministic
D. noneof these

Answer» A. deteministic

discuss

349. Evolutionary algorithms are a based approach

A. heuristic
B. metaheuristic
C. both a and b
D. noneof these

Answer» A. heuristic

discuss

350. Chromosomes are actually ?

A. line representation
B. String representa
C. Circular representat
D. all of these

Answer» B. String representa

discuss

351. Evolution Strategies is developed with

A. selection
B. mutation
C. a population of size
D. all of these

Answer» D. all of these

discuss

352. Evolution Strategies typically uses

A. real-valued vector re
B. vector representa
C. time based represe
D. none of these

Answer» A. real-valued vector re

discuss

353. Elements of ES are/is

A. Parent population siz
B. Survival populatio
C. both a and b
D. none of these

Answer» C. both a and b

discuss

354. What are different types of crossover

A. discrete and interme
B. discrete and conti
C. continuous and inte
D. none of these

Answer» A. discrete and interme

discuss

355. Determining the duration of the simulation occurs before the model is v

A. TRUE
B. FALSE

Answer» B. FALSE

discuss

356. cannot easily be transferred from one problem domain to another

A. optimal solution
B. analytical solution
C. simulation solutuon
D. none of these

Answer» C. simulation solutuon

discuss

388.	In Evolutionary programming,
A.	Individuals are represented by real- valued vector
B.	Individual solution is represented as a Finite State Machine
C.	Individuals are represented as binary string
D.	none of the mentioned

Answer» B. Individual solution is represented as a Finite State Machine

discuss

389.	In Evolutionary Strategy,
A.	Individuals are represented by real- valued vector
B.	Individual solution is represented as a Finite State Machine
C.	Individuals are represented as binary string
D.	none of the mentioned

Answer» A. Individuals are represented by real- valued vector

discuss

392.	Termination condition for EA
A.	mazimally allowed CPU time is elapsed
B.	total number of fitness evaluations reaches a given limit
C.	population diveristy drops under a given threshold
D.	All the mentioned

Answer» D. All the mentioned

discuss

393.	Which of the following operator is simplest selection operator?
A.	Random selection
B.	Proportional selection
C.	tournament selection
D.	none

Answer» A. Random selection

discuss

395.	Which of these emphasize of development of behavioral models?
A.	Evolutionary programming
B.	Genetic programming
C.	Genetic algorithm
D.	All the mentioned

Answer» A. Evolutionary programming

discuss

396.	Which selection strategy works with negative fitness value?
A.	Roulette wheel selection
B.	Stochastic universal sampling
C.	tournament selection
D.	Rank selection

Answer» D. Rank selection

discuss